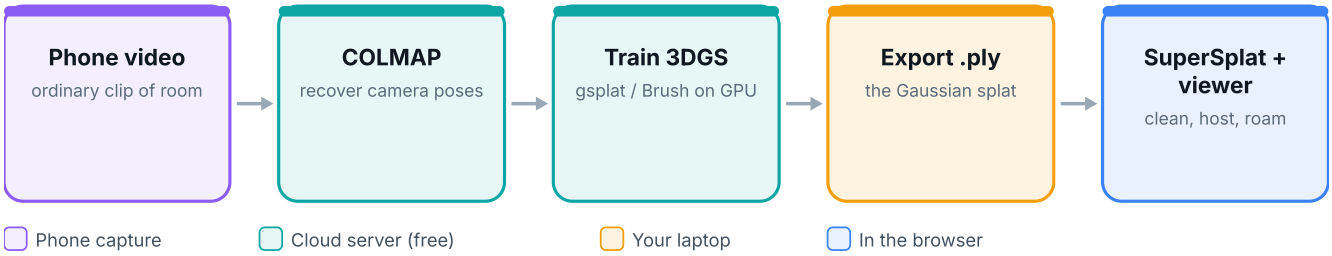
OPTION 3 · RECONSTRUCTION

3 Self-trained Gaussian Splatting

Power option

COLMAP + a 3DGS trainer on a free Colab GPU — maximum control.

In one line: do the Gaussian-splat training **yourself** on a free Colab GPU (or your laptop) from a normal video. Most control and the highest ceiling on quality — but the most technical.



COLMAP + a 3DGS trainer (gsplat / Brush / Nerfstudio) run on a Colab GPU; you tune the quality.

How it works, step by step

1. Record a steady **phone video** of the room and split it into frames.
2. **COLMAP** (structure-from-motion) recovers where each frame's camera was.
3. A **3D Gaussian Splatting trainer** (gsplat, **Brush**, or Nerfstudio's `gsplat`) optimises the splat for a few thousand iterations on a **free Colab T4 GPU**.
4. Export the trained `.ply` splat.
5. SuperSplat clean-up → our web viewer → host.

Trade-offs

✓ Strengths

- Highest possible quality — you control everything
- Works from any video; no special app
- Free on Colab; a real, defensible pipeline
- Great to describe in the Technical Abstract

✗ Weaknesses

- Most technical & time-consuming option
- COLMAP can fail on feature-poor / blurry video
- ~20–40 min per scene on Colab
- Local run is tight on a 4 GB GPU (Colab avoids this)

Tools: COLMAP, gsplat/Brush, Colab | Input: phone video | Output: .ply splat | Your compute: free Colab GPU | Cost: free

WHAT WE ACTUALLY DID

Not attempted. Earlier we looked at the related `From-video-to-3D-object` repo (COLMAP + Instant-NGP) — same idea, older engine. The modern, better-looking version is this 3DGS pipeline.

VERDICT FOR THE COMPETITION

The power option / plan B. Use it if the cloud apps' quality or free limits aren't enough, or to show deeper engineering. More effort than it's worth unless Option 1 disappoints.